

# Benchmark Toolkit Assembly Instructions

## Basic Information

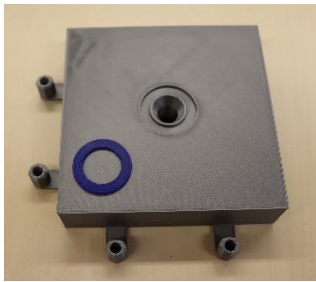
For testing and working with the toolkit in our laboratory, we have printed all parts with a Prusa i3 MK3S 3D printer. We used 1.75 mm PLA filament and the default settings of the Prusa Slicer Software if not specified otherwise in the below table:

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| <b>Print Settings:</b> | 0.2mm SPEED                                                                 |
| <b>Infill:</b>         | 10 %                                                                        |
| <b>Fill Pattern:</b>   | Grid                                                                        |
| <b>Supports:</b>       | "Everywhere" if required by the part to print (cf. below), "None" otherwise |

The .stl 3D models are designed in a way that it is possible to print all components of a benchmark toolkit part separately on with a single-color printer and glue/press-fit them together. Alternatively, you may also want to combine the component models before slicing and print them in a single pass if you can access hardware that supports multi-color printing. Below, you will find detailed assembly instructions for each part corresponding to one of the subdirectories in your benchmark\_task.zip file:

## Baseplates

### gears-BasePlate



#### Necessary components:

- 1x gears-BasePlate/0.stl in silver; this component should be printed with **supports** enabled!
- 1x gears-BasePlate/1.stl in blue
- This part may be printed in a single pass when using a multi-color 3D printer

**Assembly:** Press fit the blue ring into the circular recess on the baseplate.

### circuits-BasePlate

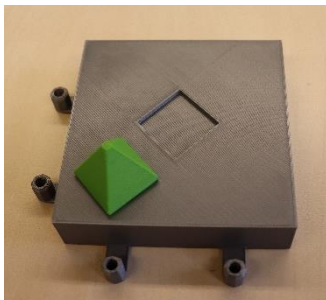


#### Necessary components:

- 1x circuits-BasePlate/0.stl in silver; this component should be printed with **supports** enabled!
- 1x circuits-BasePlate/1.stl in orange
- 1x neodymium disk magnet, 10mm x 3 mm, 2kg pull

**Assembly:** Put the magnet into the recess, then press the lid on top.

### cubes-Baseplate

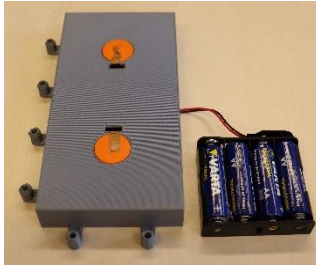


#### Necessary components:

- 1x cubes-Baseplate/0.stl in silver; this component should be printed with **supports** enabled!
- 1x cubes-Baseplate/1.stl in green

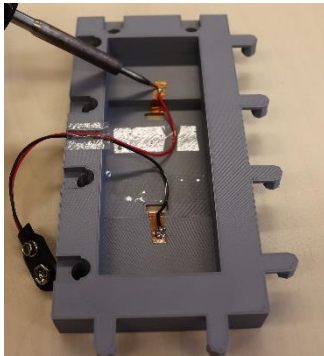
**Assembly:** Press the pyramid firmly into the recess on the top side of the baseplate.

## circuits-BaseplateBattery



### Necessary components:

- 1x circuits-BaseplateBattery/0.stl in silver; this component should be printed with **supports** enabled!
- 2x circuits-BaseplateBattery/1.stl in orange
- 2x neodymium disk magnet, 10mm x 3 mm, 2kg pull
- about 10cm of adhesive copper tape, 6mm width
- 1x wired battery case for 4 AA batteries
- 4x 1.5V AA battery

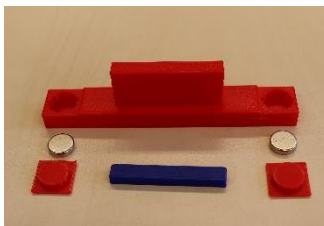


### Assembly:

- Put two magnets into the recesses and press the lids onto them.
- Wrap copper tape from the lids through the two holes to the baseplate's bottom side .
- Solder the wires of the battery case to the copper foil.

## Conductors

### circuits-ConductorBlue



### Necessary components:

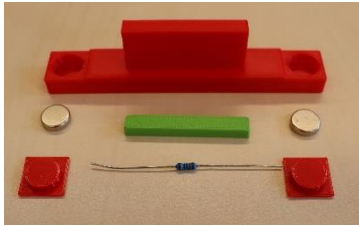
- 1x circuits-ConductorBlue/0.stl in red
- 2x circuits-ConductorBlue/1.stl in red
- 1x circuits-ConductorBlue/2.stl in blue
- 2x neodymium disk magnet, 10mm x 3 mm, 2kg pull
- About 20cm Copper Foil Tape, 6mm width



### Assembly:

- Put the magnets into the recesses on each side, then put the lids on top.
- Glue the blue marker to the conductor body.
- Wrap adhesive copper foil tape around the conductor (starting on one lid, going all around the bottom side of the conductor to the other lid).

### circuits-ConductorGreen



#### Necessary components:

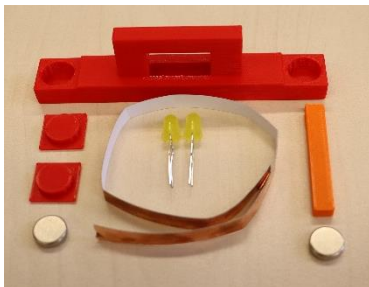
- 1x circuits-ConductorGreen/0.stl in red
- 2x circuits-ConductorGreen/1.stl in red
- 1x circuits-ConductorGreen/2.stl in green
- 2x neodymium disk magnet, 10mm x 3 mm, 2kg pull
- About 20cm of adhesive copper Tape, 6mm width
- 1x: Resistor, 150 Ohm



#### Assembly:

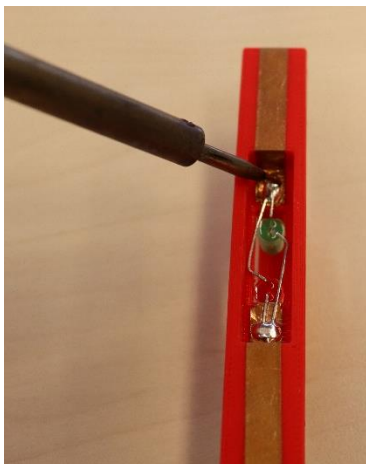
- Put the magnets into the recesses on each side, then press the lids on top.
- Glue the green marker to the main body.
- Wrap one strip of copper tape from each lid to the recess on the bottom side of the conductor.
- Solder the resistor to the ends of the two copper tape stripes to connect them.

### circuits-ConductorOrange



#### Necessary parts:

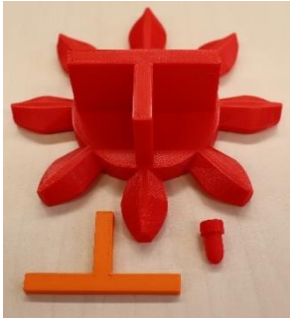
- 1x circuits-ConductorOrange/0.stl in red; this component should be printed with **supports** enabled
- 2x 1x circuits-ConductorOrange/1.stl in red
- 1x 1x circuits-ConductorOrange/2.stl in orange
- 2x neodymium disk magnet, 10mm x 3 mm, 2kg pull
- About 20cm of adhesive copper tape, 6mm width
- 2x 5mm LED, 3.3V / 20mA



#### Assembly:

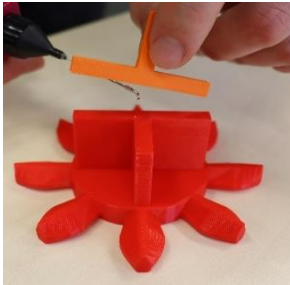
- Put the magnets into recesses on each side, then press the lids on top.
- Glue the orange marker to the conductor body.
- Wrap one strip of copper foil from each lid to the bottom side of the conductor.
- Solder the LEDs between the two copper tape strips with **reverse polarity**. This way there will always be one LED glowing, no matter which orientation the part has when built into a circuit. Both LEDs must be visible in the small "window" of the conductor body.

### gears-GearBlue, gears-GearGreen, gears-GearOrange



#### Necessary components:

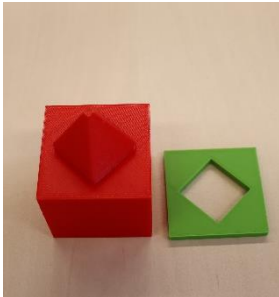
- 1x gears-GearXXXXX/0.stl in red
- 1x gears-GearXXXXX/1.stl in red
- 1x gears-GearXXXXX/2.stl in blue, green, or orange



#### Assembly:

- Glue the marker on top of the gear
- Press the pin into the hole on the bottom side of the gear.

### cubes-CubeGreen, cubes-CubeOrange



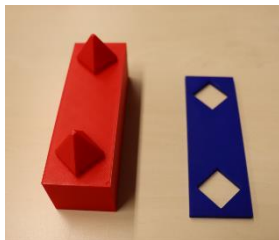
#### Necessary components:

- 1x cubes-CubeXXXXX/0.stl in red
- 1x cubes-CubeXXXXX/1.stl in green or orange

#### Assembly:

- Press the marker around the pyramid on top of the cube.

### cubes-bigCubeBlue



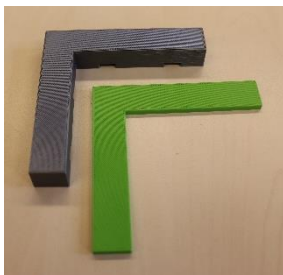
#### Necessary components:

- 1x cubes-bigCubeBlue/0.stl in red
- 1x cubes-bigCubeBlue/1.stl in blue

#### Assembly:

- Press the marker around the pyramid on top of the cube.

### general-cornerMarker



#### Necessary components:

- 1x general-cornerMarker/0.stl in silver
- 1x general-cornerMarker/1.stl in green

#### Assembly:

- Glue the green marker on top of the silver body component.